QUBO Formulation: Maximum Clique Problem

Maximum Clique Problem

Problem Statement. Given an undirected graph $G = (V, E)$ with vertex set V and edge set E, a clique is a subset $C \subseteq V$ such that every pair of distinct vertices in C is connected by an edge in E. The Maximum Clique Problem seeks to find a clique C^* of maximum cardinality, i.e., $|C^*| = \max_{C \subseteq V, C \text{ is a clique}} |C|$.

Decision Variables. For each vertex $i \in V$, introduce a binary variable $x_i \in \{0, 1\}$. The variable x_i takes the value 1 if vertex i is included in the candidate clique, and 0 otherwise. The decision vector is $\mathbf{x} = (x_i)_{i \in V} \in \{0, 1\}^{|V|}$.

QUBO Derivation. Step 1 — Write the original objective in general notation and flip the sign for minimisation. The problem requires maximising the number of selected vertices, which is equivalent to minimising its negative:

$$\min_{\mathbf{x} \in \{0,1\}^{|V|}} \left(-\sum_{i \in V} x_i \right). \tag{1}$$

Step 2 — Write each constraint in its original algebraic form using general indices. A subset of vertices forms a clique if and only if no two selected vertices are non-adjacent. For every pair of distinct vertices (i, j) that are not connected by an edge, we enforce:

$$x_i + x_j \leq 1, \quad \forall (i, j) \notin E,\ i < j. \tag{2}$$

Step 3 — Convert each constraint to a penalty term. For binary variables, the inequality $x_i + x_j \leq 1$ is violated if and only if $x_i = 1$ and $x_j = 1$. The product $x_i x_j$ equals 1 exactly on violations and 0 otherwise. We therefore encode the constraint directly as $\lambda x_i x_j$. We explicitly avoid the squared form $(x_i + x_j - 1)^2$, as it evaluates to 1 for both $(1, 1)$ and $(0, 0)$, incorrectly penalizing valid assignments. The total penalty contribution is:

$$\lambda \sum_{(i,j) \notin E,\ i<j} x_i x_j. \tag{3}$$

Step 4 — Combine objective and all penalty terms into the single QUBO Hamiltonian $H(\mathbf{x})$:

$$H(\mathbf{x}) = -\sum_{i \in V} x_i + \lambda \sum_{(i,j) \notin E,\ i<j} x_i x_j. \tag{4}$$

Step 5 — Read off the Q matrix entries and offset c from H in general closed form. Matching $H(\mathbf{x}) = \sum_{i \in V} Q_{i,i} x_i + \sum_{i<j} Q_{i,j} x_i x_j + c$, we obtain:

$$Q_{i,i} = -1, \qquad \forall i \in V, \tag{5}$$

$$Q_{i,j} = \begin{cases} \lambda & \text{if } (i,j) \notin E, \\ 0 & \text{if } (i,j) \in E, \end{cases} \qquad \forall i, j \in V,\ i < j, \tag{6}$$

$$c = 0. \tag{7}$$

Penalty Weight Justification. The objective function assigns a gain of 1 for each vertex added to the clique. A single constraint violation (selecting two non-adjacent vertices) incurs a penalty of λ. To guarantee that the global minimum corresponds to a feasible clique, the penalty must strictly exceed the maximum possible objective benefit of violating a constraint. Since the maximum single-variable objective coefficient magnitude is 1, setting $\lambda > 1$ ensures any violation outweighs the best possible objective gain from adding a vertex. We adopt $\lambda = 2$ as a robust integer choice that satisfies this condition for any instance size $n = |V|$.

Correctness Argument. Minimising $H(\mathbf{x})$ over $\{0,1\}^{|V|}$ recovers the optimal solution because (i) any feasible clique assignment satisfies all constraints, yielding a penalty of zero and an energy equal to $-|C|$; (ii) any infeasible assignment violates at least one non-edge constraint, incurring a penalty of at least $\lambda > 1$. Since the maximum possible reduction in the objective term from adding an additional vertex is 1, the penalty strictly dominates any infeasible configuration's potential energy advantage. Consequently, the global minimum of H must be attained at a feasible assignment with the largest possible cardinality, i.e., the maximum clique.

Illustrative Example. Consider a graph with $n = 3$ vertices $\{0, 1, 2\}$ and no edges ($E = \emptyset$). All pairs are non-adjacent, so the constraint $x_i + x_j \leq 1$ applies to $(0, 1)$, $(0, 2)$, and $(1, 2)$. Using $\lambda = 2$, the general formulas instantiate to:

$$H(x_0, x_1, x_2) = -x_0 - x_1 - x_2 + 2x_0x_1 + 2x_0x_2 + 2x_1x_2. \tag{8}$$

The concrete Q matrix entries are $Q_{0,0} = Q_{1,1} = Q_{2,2} = -1$, $Q_{0,1} = Q_{0,2} = Q_{1,2} = 2$, and $c = 0$. Evaluating all $2^3 = 8$ binary assignments, the minimum energy is -1, achieved at $(1, 0, 0)$, $(0, 1, 0)$, and $(0, 0, 1)$. These correspond to cliques of size 1, which is optimal for an edgeless graph.